

Rule 1 — Hard hat required

Critical



helmet

Compliant — helmet worn

Hard hat present and positioned correctly on head. No violation raised.



no helmet

Violation — no hard hat

Worker detected with no helmet. `no\_hard\_hat` class matched to person box. Violation raised.



conf: 0.38
uncertain

Uncertain — low confidence

Helmet detected but confidence < 0.50. Possibly occluded or distant. Flagged SAFE-UNCERTAIN.

Rule 2 — High-visibility vest required

Critical



vest on

Compliant — vest worn

High-vis vest visible on torso. `safety\_vest` class detected. No violation raised.



no vest

Violation — no safety vest

Worker in plain clothing in active work zone. `no\_safety\_vest` class matched to person box.



vest open

Violation — vest open / partial

Vest present but hanging open — not providing adequate visibility. Treated as no-vest violation.

Rule 3 — Uncertainty flagging

Review trigger



distant worker
conf: 0.31
flag for review

Distant / small worker



occluded

Partially occluded



low light

Low-light / shadow